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 Studierichting: Informatica  
 Oefeningenreeks 5A nr. 5.7

$$\begin{aligned}
 BI \int_{\pi}^{2\pi} \cos^2(x) dx &\Rightarrow OBI = \int \cos^2(x) dx \\
 &= \int \frac{1 + \cos(2x)}{2} dx = \frac{1}{2} \int 1 + \cos(2x) dx = \frac{1}{2} \left[ \int 1 dx + \int \cos(2x) \right] \\
 \text{Stel: } t = 2x &\Rightarrow dt = 2dx \Leftrightarrow dx = \frac{dt}{2} \\
 &= \frac{1}{2} \left[ x + c + \frac{1}{2} \int \cos t dt \right] = \frac{1}{2} \left[ x + c + \frac{1}{2} \sin 2x + c \right] = \frac{1}{2} x + \frac{1}{4} \sin(2x) + c \\
 BI &= \left[ \frac{1}{2} x + \frac{1}{4} \sin(2x) \right]_{\pi}^{2\pi} = \frac{1}{2} (2\pi) + \frac{1}{4} \sin(2 \cdot 2\pi) - \frac{1}{2} (\pi) - \frac{1}{4} \sin(2 \cdot \pi)
 \end{aligned}$$

Doordat  $\sin(k \cdot 2\pi) = 0$ , met  $k \in \mathbb{Z}$

$$\frac{1}{2} \left( 2\pi - \pi + \frac{1}{2} \sin(4\pi) - \frac{1}{2} \sin(4\pi) \right) = \frac{\pi}{2}$$

## References